

Celebal Excellence Internship

[Week – 2 Task – SQL Sales Data Analysis]

Objective: Analyze sales data using SQL with filtering, aggregation, and business queries. Steps: 1. Load dataset into a SQL database. 2.Explore table (schema, sample data). 3.Apply WHERE filters (region, category, date, sales). 4.Use GROUP BY for aggregations (sales, quantity, averages). 5.Sort and limit results (top products, top categories). 6.Solve use cases (monthly trends, top customers, duplicates). 7.Validate results (row counts, data quality).

Solution:

1. Load and Explore Dataset

The dataset is imported in SQL server and explored using some SQL queries:

- Displaying top 5 records

Query:

```
SELECT TOP 5 *  
FROM SUPERSTORE;
```

Result:

Results Messages																	
Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name	
1	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BQ-10001798	Furniture	Bookcases	Bush Somerset
2	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fab
3	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DI-13045	Darin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Ac
4	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR450C
5	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000780	Office Supplies	Storage	Eldon Fold N R

	Department	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name	Sales	Quantity	Discount	Profit
1	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.959991455078	2	0	41.9136009216309
2	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, R...	731.940002441406	3	0	219.582000732422
3	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Uni...	14.6199998855591	2	0	6.87139987945557
4	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.577514648438	5	0.449999988079071	-383.031005859375
5	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.3680000305176	2	0.200000002980232	2.51640009880066

- Displaying count of rows

Query:

```
SELECT COUNT(*) AS TotalRows
```

```
FROM SUPERSTORE;
```

Result:

	TotalRows
1	9994

- Displaying distinct categories

Query:

```
SELECT DISTINCT Category
```

```
FROM SUPERSTORE;
```

Result:

	Category
1	Office Supplies
2	Furniture
3	Technology

- Displaying distinct regions

Query:

```
SELECT DISTINCT Region
```

```
FROM SUPERSTORE;
```

Result:

	Region
1	East
2	South
3	West
4	Central

2. Apply WHERE Filters:

- Displaying records with west region

Query:

```
SELECT TOP 5 *  
FROM SUPERSTORE  
WHERE Region ='West';
```

Result:

Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name	
1	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by Univ...
2	6	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessor...
3	7	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322
4	8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone
5	9	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BB-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by Sam...

Product_Name	Sales	Quantity	Discount	Profit
Self-Adhesive Address Labels for Typewriters by Univ...	14.6199998855591	2	0	6.87139987945557
Eldon Expressions Wood and Plastic Desk Accessor...	48.8600006103516	7	0	14.1694002151489
Newell 322	7.28000020980835	4	0	1.96560001373291
Mitel 5320 IP Phone VoIP phone	907.151977539063	6	0.200000002980232	90.715202331543
DXL Angle-View Binders with Locking Rings by Sam...	18.503999710083	3	0.200000002980232	5.78249979019165

- Displaying records in technology category

Query:

```
SELECT TOP 5 *  
FROM SUPERSTORE  
WHERE CATEGORY = 'Technology';
```

Result:

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country
1	8	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
2	12	CA-2014-115812	2014-06-09	2014-06-14	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
3	20	CA-2014-143336	2014-08-27	2014-09-01	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States
4	27	CA-2016-121755	2016-01-16	2016-01-20	Second Class	EH-13945	Eric Hoffmann	Consumer	United States
5	36	CA-2016-117590	2016-12-08	2016-12-10	First Class	GH-14485	Gene Hale	Corporate	United States

City	State	Postal_Code	Region	Product_ID	Category	Sub_Category
Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones
Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones
San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones
Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Accessories
Richardson	Texas	75080	Central	TEC-PH-10004977	Technology	Phones

Product_Name	Sales	Quantity	Discount	Profit
Mitel 5320 IP Phone VoIP phone	907.151977539063	6	0.200000002980232	90.715202331543
Konftel 250 Conference phone - Charcoal black	911.424011230469	4	0.200000002980232	68.3567962646484
Cisco SPA 501G IP Phone	213.479995727539	3	0.200000002980232	16.0109996795654
Imation 8GB Mini TravelDrive USB 2.0 Flash Drive	90.5699996948242	3	0	11.7741003036499
GE 30524EE4	1097.5439453125	7	0.200000002980232	123.473701477051

- Displaying high scale entries

Query:

```
SELECT TOP 5 *
FROM SUPERSTORE
WHERE SALES > 500;
```

Result:

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country
1	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States
2	4	US-2015-1089...	2015-10-11	2015-10-18	Standard Cl...	SO-20335	Sean O'Donnell	Consumer	United States
3	8	CA-2014-115812	2014-06-09	2014-06-14	Standard Cl...	BH-11710	Brosina Hoffman	Consumer	United States
4	11	CA-2014-115812	2014-06-09	2014-06-14	Standard Cl...	BH-11710	Brosina Hoffman	Consumer	United States
5	12	CA-2014-115812	2014-06-09	2014-06-14	Standard Cl...	BH-11710	Brosina Hoffman	Consumer	United States

	City	State	Postal_Code	Region	Product_ID	Category	Sub_Category
1	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs
2	Fort Laud...	Florida	33311	South	FUR-TA-10000577	Furniture	Tables
3	Los Ange...	Califor...	90032	West	TEC-PH-10002275	Techno...	Phones
4	Los Ange...	Califor...	90032	West	FUR-TA-10001539	Furniture	Tables
5	Los Ange...	Califor...	90032	West	TEC-PH-10002033	Techno...	Phones

Product_Name	Sales	Quantity	Discount	Profit
Hon Deluxe Fabric Upholstered Stacking Chairs, R...	731.940002441406	3	0	219.582000732422
Bretford CR4500 Series Slim Rectangular Table	957.577514648438	5	0.4499...	-383.0310058593...
Mitel 5320 IP Phone VoIP phone	907.151977539063	6	0.2000...	90.715202331543
Chromcraft Rectangular Conference Tables	1706.18395996094	9	0.2000...	85.309196472168
Konftel 250 Conference phone - Charcoal black	911.424011230469	4	0.2000...	68.3567962646484

- Displaying orders from year 2017

Query:

```
SELECT TOP 5 *
FROM SUPERSTORE
WHERE ORDER_DATE LIKE '2017%';
```

Result:

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country
1	13	CA-2017-114412	2017-04-15	2017-04-20	Standard Class	AA-10480	Andrew Allen	Consumer	United States
2	24	US-2017-1569...	2017-07-16	2017-07-18	Second Class	SF-20065	Sandra Flanagan	Consumer	United States
3	35	CA-2017-107727	2017-10-19	2017-10-23	Second Class	MA-17560	Matt Abelman	Home Of...	United States
4	42	CA-2017-120999	2017-09-10	2017-09-15	Standard Class	LC-16930	Linda Cazamias	Corporate	United States
5	44	CA-2017-139619	2017-09-19	2017-09-23	Standard Class	ES-14080	Erin Smith	Corporate	United States

City	State	Postal_Code	Region	Product_ID	Category	Sub_Category	Product_Name
Concord	North Carolina	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967
Philad...	Pennsylvania	19140	East	FUR-CH-100027...	Furniture	Chairs	Global Delux...
Houston	Texas	77095	Central	OFF-PA-10000249	Office Supplies	Paper	Easy-staple p...
Naper...	Illinois	60540	Central	TEC-PH-100040...	Technology	Phones	Panasonic Kx...
Melbo...	Florida	32935	South	OFF-ST-10003282	Office Supplies	Storage	Advantus 10-...

Sales	Quantity	Discount	Profit
15.5520000457764	3	0.200000002980232	5.44320011138916
71.3720016479492	2	0.300000011920929	-1.019600033760...
29.4720001220703	3	0.200000002980232	9.94680023193359
147.167999267578	4	0.200000002980232	16.5564002990723
95.6159973144531	2	0.200000002980232	9.56159973144531

3. Use GROUP BY aggregation

- Displaying total sales by region

Query:

```
SELECT REGION, SUM(SALES) AS TotalSales
FROM SUPERSTORE
GROUP BY REGION
ORDER BY TotalSales DESC;
```

Result:

	REGION	TotalSales
1	West	725457.823114276
2	East	678781.237776518
3	Central	501239.889593184
4	South	391721.905365348

- Displaying total sales by Category

Query:

```
SELECT CATEGORY, SUM(SALES) AS TotalSales
FROM SUPERSTORE
GROUP BY CATEGORY
ORDER BY TotalSales DESC;
```

Result:

	CATEGORY	TotalSales
1	Technology	836154.030862212
2	Furniture	741999.793703437
3	Office Supplies	719047.031283677

- Displaying total quantity sold by Category

Query:

```
SELECT CATEGORY, SUM(QUANTITY) AS TotalQuantity
FROM SUPERSTORE
GROUP BY CATEGORY
ORDER BY TotalQuantity DESC;
```

Result:

	CATEGORY	TotalQuantity
1	Office Supplies	22906
2	Furniture	8028
3	Technology	6939

- Displaying average profit by Category

Query:

```
SELECT CATEGORY, AVG(PROFIT) AS AverageProfit
FROM SUPERSTORE
GROUP BY CATEGORY
ORDER BY AverageProfit;
```

Result:

	CATEGORY	AverageProfit
1	Furniture	8.90154373143923
2	Office Supplies	20.3270495597804
3	Technology	78.7520018695341

4. Sort and limit results

- Displaying top 10 products by sales

Query:

```
SELECT TOP 10 [PRODUCT_NAME] , SUM(SALES) AS
TotalSales
FROM SUPERSTORE
GROUP BY [PRODUCT_NAME]
ORDER BY TotalSales DESC ;
```

Result:

	PRODUCT_NAME	TotalSales
1	Canon imageCLASS 2200 Advanced Copier	61599.822265625
2	Fellowes PB500 Electric Punch Plastic Comb Binding ...	27453.3840332031
3	Cisco TelePresence System EX90 Videoconferencing ...	22638.48046875
4	HON 5400 Series Task Chairs for Big and Tall	21870.5755004883
5	GBC DocuBind TL300 Electric Binding System	19823.478515625
6	GBC Ibimaster 500 Manual ProClick Binding System	19024.5002441406
7	Hewlett Packard LaserJet 3310 Copier	18839.6859130859
8	HP Designjet T520 Inkjet Large Format Printer - 24" C...	18374.8952636719
9	GBC DocuBind P400 Electric Binding System	17965.0678710938
10	High Speed Automatic Electric Letter Opener	17030.3117675781

- Displaying top 10 categories by sales

Query:

```
SELECT CATEGORY, SUM(SALES) AS TotalSales
FROM SUPERSTORE
GROUP BY CATEGORY
ORDER BY TotalSales DESC;
```

Result:

	CATEGORY	TotalSales
1	Technology	836154.030862212
2	Furniture	741999.793703437
3	Office Supplies	719047.031283677

- Displaying top 10 customers by sales

Query:

```
SELECT TOP 10 CUSTOMER_NAME, SUM(SALES) AS
TotalSales
```



```

FROM SUPERSTORE
GROUP BY CUSTOMER_NAME
ORDER BY TotalSales;

```

Result:

	CUSTOMER_NAME	TotalSales
1	Thais Sissman	4.83300018310547
2	Lela Donovan	5.30399990081787
3	Carl Jackson	16.5200004577637
4	Mitch Gastineau	16.7389998435974
5	Roy Skaria	22.3280000686646
6	Susan Gilcrest	47.946001291275
7	Ricardo Emerson	48.3600006103516
8	Larry Blacks	50.1879988908768
9	Adrian Shami	58.8199982643127
10	Jasper Cacioppo	71.2629990577698

5. Solve use cases

- Displaying Monthly sales trend

Query:

```

SELECT
YEAR(ORDER_DATE) AS OrderYear,
MONTH(ORDER_DATE) AS OrderMonth,
SUM(SALES) AS TotalSales
FROM SUPERSTORE
GROUP BY YEAR(ORDER_DATE), MONTH(ORDER_DATE)
ORDER BY OrderYear , OrderMonth;

```

Result:

	OrderYear	OrderMonth	TotalSales
1	2014	1	14236.8949842453
2	2014	2	4519.89195883274
3	2014	3	55691.0094711781
4	2014	4	28295.344894886
5	2014	5	23648.2868702412
6	2014	6	34595.1274838448
7	2014	7	33946.3927909136
8	2014	8	27909.4681773186

	OrderYear	OrderMonth	TotalSales
9	2014	9	81777.3513105512
10	2014	10	31453.3927481174
11	2014	11	78628.7165292501
12	2014	12	69545.6202806234
13	2015	1	18174.0755178928
14	2015	2	11951.4109165668
15	2015	3	38726.25218153
16	2015	4	34195.2084971666

OrderYear	OrderMonth	TotalSales
2016	1	18542.4910029173
2016	2	22978.8151884079
2016	3	51715.8751164675
2016	4	38750.0386896133
2016	5	56987.7272846699
2016	6	40344.5338503718
2016	7	39261.9630224705
2016	8	31115.3742319345

OrderYear	OrderMonth	TotalSales
2017	5	44261.1099559069
2017	6	52981.725668788
2017	7	45264.415922761
2017	8	63120.8877636194
2017	9	87866.6517419815
2017	10	77776.9230260849
2017	11	118447.824670434
2017	12	83829.318554163

- Displaying monthly count of orders

Query:

```
SELECT
YEAR(ORDER_DATE) AS OrderYear,
MONTH(ORDER_DATE) AS OrderMonth,
COUNT(DISTINCT ORDER_ID) AS TotalOrders
FROM SUPERSTORE
GROUP BY YEAR(ORDER_DATE), MONTH(ORDER_DATE)
ORDER BY OrderYear, OrderMonth;
```

Result:

OrderYear	OrderMonth	TotalOrders
2014	1	32
2014	2	28
2014	3	71
2014	4	66
2014	5	69
2014	6	66
2014	7	65
2014	8	72

OrderYear	OrderMonth	TotalOrders
2015	1	29
2015	2	36
2015	3	79
2015	4	72
2015	5	74
2015	6	68
2015	7	66
2015	8	68

OrderYear	OrderMonth	TotalOrders
2015	12	161
2016	1	48
2016	2	45
2016	3	86
2016	4	89
2016	5	108
2016	6	97
2016	7	96

OrderYear	OrderMonth	TotalOrders
2017	5	118
2017	6	133
2017	7	111
2017	8	111
2017	9	226
2017	10	147
2017	11	261
2017	12	224

- Displaying top 10 customers by total sales

Query:

```
SELECT TOP 10
CUSTOMER_NAME ,
SUM(SALES) AS TotalSales
FROM SUPERSTORE
GROUP BY CUSTOMER_NAME
ORDER BY TotalSales DESC;
```

Result:

	CUSTOMER_NAME	TotalSales
1	Sean Miller	25043.0504860878
2	Tamara Chand	19052.217195034
3	Raymond Buch	15117.3389587402
4	Tom Ashbrook	14595.6198644638
5	Adrian Barton	14473.571243763
6	Ken Lonsdale	14175.2289050817
7	Sanjit Chand	14142.3341379166
8	Hunter Lopez	12873.297662735

9	Sanjit Engle	12209.4381403923
10	Christopher Conant	12129.0715373755

- Displaying duplicate order ID's

Query:

```
SELECT
ORDER_ID,
COUNT(*)AS DuplicateCount
FROM SUPERSTORE
GROUP BY ORDER_ID
HAVING COUNT(*)>1
ORDER BY DuplicateCount DESC;
```

Result:

	ORDER_ID	DuplicateCount
1	CA-2017-100111	14
2	CA-2017-157987	12
3	CA-2016-165330	11
4	US-2016-108504	11
5	US-2015-126977	10
6	CA-2016-105732	10
7	CA-2015-131338	10
8	US-2016-114013	9
9	CA-2016-145177	9
10	US-2015-163433	9
11	CA-2014-106439	9
12	CA-2015-104346	9
13	CA-2015-132626	9
14	CA-2015-164882	9
15	CA-2017-140949	9
16	CA-2017-117457	9

	ORDER_ID	DuplicateCount
24...	CA-2014-166191	2
24...	CA-2017-135307	2
24...	CA-2017-159506	2
24...	CA-2015-142993	2
24...	CA-2016-101651	2
24...	CA-2017-145884	2
24...	CA-2016-165218	2
24...	CA-2014-160773	2
24...	US-2017-121251	2
24...	CA-2015-123113	2
24...	CA-2017-154011	2
24...	CA-2016-146941	2
24...	CA-2014-151001	2
24...	CA-2015-121132	2
24...	CA-2016-131380	2
24...	US-2014-131870	2

- Displaying top 10 states by sales

Query:

```
SELECT TOP 10
STATE,
SUM(SALES) AS TotalSales
FROM SUPERSTORE
GROUP BY STATE
```

ORDER BY TotalSales DESC

Result:

	STATE	TotalSales
1	California	457687.630218506
2	New York	310876.270583153
3	Texas	170188.045670152
4	Washin...	138641.269838929
5	Pennsyl...	116511.912908077
6	Florida	89473.7084289789
7	Illinois	80166.1008920074
8	Ohio	78258.1357239485
9	Michigan	76269.6139240265
10	Virginia	70636.7199046612

- Displaying top 10 sub categories by sales

Query:

```
SELECT TOP 10  
SUB_CATEGORY ,  
SUM(SALES) AS TotalSales  
FROM SUPERSTORE  
GROUP BY SUB_CATEGORY  
ORDER BY TotalSales DESC;
```

Result:

	SUB_CATEGORY	TotalSales
1	Phones	330007.054325342
2	Chairs	328449.102329254
3	Storage	223843.608764172
4	Tables	206965.531606674
5	Binders	203412.732438266
6	Machines	189238.631017685
7	Accessories	167380.317870259
8	Copiers	149528.027648926
9	Bookcases	114879.996025085
10	Appliances	107532.161303163

